



An orientation estimation strategy for low cost IMU using a nonlinear Luenberger observer

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ABSTRACT

In this work a new orientation estimation strategy for a non-accelerated platform is presented. The proposed strategy is based on a low-cost inertial measurement unit (IMU). Orientation angles are obtained using a nonlinear Luenberger observer, while the common offset issues on the magnetometer are calibrated by a recursive least-square algorithm. The strategy is implemented using a TM4C123GXL microcontroller and a MPU9150 sensor (nine-axis: gyro, accelerometer and magnetometer), which makes it a low-cost, compact system. Experimental results for the calibration and estimation procedures are shown to validate the proposal.

1. Introduction

There are many applications in which it is needed to know the orientation angles of a rigid mobile platform, for example, attitude estimation in aerial [1] and ground vehicles [2], stabilization platforms for cameras, human body orientation systems [3] and industrial machines applications [4], among others.

The most popular approaches to obtain the orientation angles are based on inertial sensors, due to the availability of accurate low-cost inertial sensors which makes possible to obtain an angular position estimator that can be implemented in a low cost and compact device. However, the computational burden of the estimation strategy must not be very high, so as to program it into a low cost microcontroller.

From a theoretical point of view, rotation matrices are elements of a Lie group, the so-called special orthogonal group $SO(3)$. Its Lie algebra is the space of skew-symmetric matrices which is isomorphic to $\mathbb{R}^3$. Such vectors represent angular velocities (spatial or material) in the field of rotational dynamics [5]. It implies that, if the time history of the angular velocity is known (e.g., from gyroscope measurements), then the attitude history can be obtained by numerical integration. In addition, there are several ways to parameterize the $SO(3)$ group, some being preferable over others. Three-dimensional parameterizations, such as Euler's angles, the Cartesian rotation vector, and Cayley's parameters have serious drawbacks, since they cannot

be global and singular free at the same time [6]. Furthermore, Euler's angle-based parameterizations are not computationally efficient in describing large arbitrary finite rotations (spacecraft dynamics, vehicle dynamics, robotics, etc.) due to the trigonometric nature of the components of the resulting rotation matrix. An often used alternative to three-dimensional-based parameterizations is the well-known four-dimensional generalization of complex numbers called quaternions. However, handling rotations via quaternions require enforcing a unit norm constraint (unit quaternions). Despite this additional requirement, they allow us to represent the most general possible motion of a body, while at the same time producing rotation matrices with polynomial components. Unit quaternions can also be easily restated in terms of Euler's parameters or Rodriguez's parameters [7].

However, the gyroscope drift (offset that slowly moves in time) [8] is a known problem in the literature, therefore attitude cannot be obtained by numerical integration in practice applications. For this reason, several approaches are based on statistical techniques, such as Kalman filters (KF) or extended Kalman filters (EKF). Some of them are discussed below.

In [9] two similar estimation strategies are presented: the first one precalculates a quaternion vector using the algorithm QUaternion ESTimator (QUEST) from the accelerometer and magnetometer measurements. The information obtained from the gyro and a KF are used

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to improve the estimation of the precalculated quaternion. The second one combines all the measurements using an extended Kalman filter. The first proposal has the advantage that the filter is applied to a linear dynamic system, therefore the computational cost is lower than for the EKF case.

In [10], the estimation is carried out using a direct integration of the angular velocity vector and then corrected by an EKF using measurements from the accelerometer and the magnetometer in two steps. This reduces the computational burden compared with only one EKF that combines all the measurements.

The author in [11] proposes a quaternion-based sensor fusion approach for orientation estimation using a low complexity linear Kalman filter based on the differential state equation, which is improved by compensating for the sample-wise gyroscope drift in the time update system instead of the measurement update system.

Recently, an unscented Kalman filter with multiple models and a decision tree for attitude estimation was proposed in [12], which improves the estimation error with respect to previous proposals and [13] uses a KF and a gradient descent algorithm that is more robust against magnetic disturbances compared to previous KF based proposals.

Filter and complementary observers approach can be found in the bibliography as an alternative to overcome the global stability guarantee limitation of the extended Kalman filter. In [14,15] a filter based on a descending gradient algorithm is presented. This filter is based in a 9 axis IMU and a quaternion-based representation. It is computationally more efficient with respect to proposals based on KF, and it also adds compensation for magnetic distortions. The proposal in [16] incorporates a motion kinematic model and adopts a two-layer filter architecture avoiding the intermediate step of bias correction of the gyros, another well known problem for orientation estimation. On the other hand, the strategy [17] is more robust to magnetic disturbances than previously proposed filters. [18] is based on [15], but the convergence rate is improved and it presents a greater immunity against distortions in magnetic measurements. The proposal in [19] is similar to the filter presented by [15], but it reduces the mathematical complexity and therefore the run time.

In [20] a deep learning-based framework is presented as a new alternative to the proposals based on KF and complementary filters, which could improve the estimation error in some circumstances with respect to complementary filters. However, the authors does not include implementation results in a microcontroller neither an analysis of the required processing capacity.

There are cases in which it is desired to estimate the orientation of a platform in the presence of linear accelerations. These accelerations introduce errors in the IMU based orientation estimation strategies. In [21] a KF is used to estimate the orientation and the external acceleration is determined from the filter residue. In [3] the conventional KF approach is modified considering the direction of linear acceleration, which help to prevent this from affecting the performance of the orientation estimation.

Another well known problem for orientation estimation is produced by gyros bias. To avoid gyros bias, the device can be initially calibrated or either the bias drift can be compensated by the estimation strategy. Accelerometers usually have no major bias problems, but sometimes their gain requires a previous calibration. Magnetometers however, have well known offset problems and requires a calibration procedure when they are used for orientation estimation [22]. This is due to magnetometers measure the Earth's magnetic field, which is very weak, and then it is prone to be altered by other nearby magnetic fields. These magnetometer offsets are classified into two categories, Hard Iron Offset (HIO) and Soft Iron Offset (SIO) [23]. HIO correspond to the influence of ferromagnetic components located on the electronic board where the magnetometer is mounted. SIO is produced by the temporary magnetization of near metallic components due to the Earth's magnetic field effect. The HIO does not depend on the electronic board orientation, so it is relatively easy to obtain its value. It is usually the

most notable of the two offset effects. Once this value is known, it can be subtracted as a constant offset from the measurements of the magnetometer.

There are different methods for magnetometer calibration. Some of them require the addition of external instruments, which are not practical in low-cost applications, such as the calibration platform shown in [24]. For this reason, several works have emerged which only used IMU measurements in order to carry out the calibration. Magnetometer measurements can be used to calibrate it with a method known as ellipsoid fitting [25,26] to a sphere centered at the origin of coordinates. The disadvantage of this method is that sufficient data must be available for the least squares method to converge. Alternatively, magnetometer calibration methods using inertial sensors were also proposed [27]. This method is formulated as the solution to a maximum likelihood problem. It is done offline and it also requires the sensor to be moved in all the directions for the calibration to be performed. The authors in [28] used an unscented Kalman filter, requiring fewer movements to calibrate the magnetometer.

The main contribution of this work is a new orientation estimation strategy based on a nonlinear observer. The main feature of this proposal is that the nonlinear observer estimates only four state variables and it does not need the real time calculation of inverse matrices. This decreases the computational burden with respect to the proposals reviewed above.

As a secondary contribution, in this work a recursive algorithm for HIO estimation and the orientation observer are simultaneously implemented in a low-cost device. Even though HIO calibration is essential to use the magnetometer, in most of the reviewed works, orientation an calibrations are problems that are analyzed separately. Instead, the present proposal allows the calibration to be adapted to the slow variations of HIO that may be happening during the operation, also considering the computational burden. In this paper, the implemented estimation algorithm is based on the ellipsoid fitting idea [25,26], which was simplified to estimate only the HIO.

The observer and the HIO recursive estimator were implemented in a low-cost Cortex ARM4 microcontroller. Experimental results obtained within a laboratory prototype validate the proposed theoretical procedures. In addition, it was verified that the required processing time of the microcontroller to execute both procedures is less than 5 percent of its capacity.

2. Theoretical background

2.1. Coordinates systems

In this paper, two reference systems are used to describe the orientation of a body (dynamic platform) in the three-dimensional space: (i) a Newtonian or inertial system $N(x, y, z)$, whose x axis is north-oriented and z axis is pointing down; and (ii) a body-fixed reference-frame $B(\xi, \eta, \zeta)$. It should be mentioned that the dynamic system discussed here does not undergo any translational motion. i.e., its center of mass (mc) remains fixed in space and coincides with the origin o of the inertial frame N (Fig. 1). The body orientation is achieved by means of a parameterization based on unitary quaternions, also known as Euler parameters. Although the quaternion form a four-dimensional vector space, they are free of some of the deficiencies of other commonly used angular coordinates, such as the phenomenon of gimbal lock in the so-called Euler angles [29].

2.2. Quaternion description

According to Euler's rotation theorem, any rotation in three dimensions can be represented as a combination of a rotation axis $\hat{n}$ and a scalar φ . In light of this, unit quaternions give a simple way to encode this axis-angle representation in four parameters, which can be used to

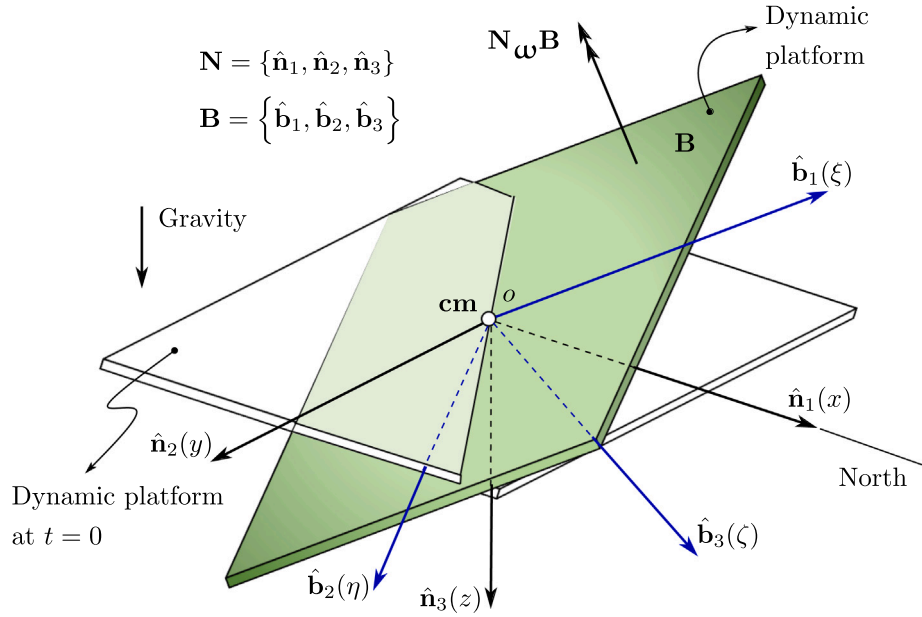


Fig. 1. Definition of the coordinate systems.

describe the orientation of a body relative to the origin of $\mathbb{R}^3$ [30]. Let us consider a unit quaternion defined by,

$$\mathbf{q}(t) = [q_0(t), q_1(t), q_2(t), q_3(t)]^T \quad (1)$$

where $q_i(t)$ are the Euler parameters, which are subject to the following constraint,

$$\mathbf{q}^T \mathbf{q} - 1 = 0. \quad (2)$$

The quaternion expressed in (1) can be algebraically manipulated into a quaternion-derived rotation matrix as follows,

$$\mathbf{T}_{\mathbf{NB}} = 2 \begin{bmatrix} q_0^2 + q_1^2 - \frac{1}{2} & q_1 q_2 - q_0 q_3 & q_1 q_3 + q_0 q_2 \\ q_1 q_2 + q_0 q_3 & q_0^2 + q_2^2 - \frac{1}{2} & q_2 q_3 - q_0 q_1 \\ q_1 q_3 - q_0 q_2 & q_2 q_3 + q_0 q_1 & q_0^2 + q_3^2 - \frac{1}{2} \end{bmatrix}. \quad (3)$$

where $\mathbf{T}_{\mathbf{NB}} : \mathbf{B} \rightarrow \mathbf{N} \in SO(3)$ represents the rotation matrix from the body-fixed frame $\mathbf{B}$ to the inertial frame $\mathbf{N}$, and $SO(3)$ is the so-called special orthogonal group. This rotation matrix is later used in this work to express variables on different frames.

The components of the angular velocity vector (obtained from the gyroscope measurements) are related to the quaternion-based rotational parameters by means of the following classical expression [29],

$${}^{\mathbf{N}}\boldsymbol{\omega}^{\mathbf{B}} = 2\mathbf{L}\dot{\mathbf{q}}, \quad (4)$$

where the dot represents time derivative; ${}^{\mathbf{N}}\boldsymbol{\omega}^{\mathbf{B}}$ is the angular velocity vector of $\mathbf{B}$ relative to the inertial frame $\mathbf{N}$ and spanned in the body-fixed basis, $\mathbf{B}$; and $\mathbf{L}$ is a matrix defined by,

$$\mathbf{L} = \begin{bmatrix} -q_1 & q_0 & q_3 & -q_2 \\ -q_2 & -q_3 & q_0 & q_1 \\ -q_3 & q_2 & -q_1 & q_0 \end{bmatrix}. \quad (5)$$

After some algebraic manipulations, (4) can be rewritten as follows,

$$\dot{\mathbf{q}} = \frac{1}{2} \tilde{\boldsymbol{\omega}}_{(t)} \mathbf{q}, \quad (6)$$

where $\tilde{\boldsymbol{\omega}}_{(t)}$ is a skew-symmetric matrix given by,

$$\tilde{\boldsymbol{\omega}}_{(t)} = \begin{bmatrix} 0 & -\omega_\xi & -\omega_\eta & -\omega_\zeta \\ \omega_\xi & 0 & \omega_\zeta & -\omega_\eta \\ \omega_\eta & -\omega_\zeta & 0 & \omega_\xi \\ \omega_\zeta & \omega_\eta & -\omega_\xi & 0 \end{bmatrix}. \quad (7)$$

and $\omega_\xi, \omega_\eta, \omega_\zeta$ are the components of the angular velocity vector expressed in the body-fixed basis, $\mathbf{B}$, measured by the gyro.

2.3. System outputs

The accelerometer and magnetometer measurements can be expressed as a function of the state variables $\mathbf{q}$ and defined as the outputs of the system (6). Next, the mathematical procedure to obtain the outputs as a function of $\mathbf{q}$ is detailed.

In the case that the platform is aligned with the inertial frame $\mathbf{N}$ (fixed to the Earth surface) and oriented along the Newtonian coordinates x and y , then the acceleration vector given by the sensor in this position is given by,

$$\mathbf{a}_{\mathbf{N}} = \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}, \quad (8)$$

where $\mathbf{a}_{\mathbf{N}}$ is the acceleration vector expressed in the inertial frame $\mathbf{N}$, and g is the acceleration due to the gravitational field.

If the platform rotates around the point O , the measurements given by the accelerometer will contain gravity components in the three axes. By using (3) along with (8), the acceleration vector solved in the body-fixed frame $\mathbf{B}$ can be written as follows,

$$\mathbf{a}_{\mathbf{B}} = \begin{bmatrix} a_\xi \\ a_\eta \\ a_\zeta \end{bmatrix} = \mathbf{T}_{\mathbf{BN}} \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} = \begin{bmatrix} 2q_1 q_3 - 2q_0 q_2 \\ 2q_0 q_1 + 2q_2 q_3 \\ 2q_0^2 + 2q_3^2 - 1 \end{bmatrix} g, \quad (9)$$

where $\mathbf{T}_{\mathbf{BN}} = (\mathbf{T}_{\mathbf{NB}})^T$.

In the same way, in the case that the magnetometer (mounted on the platform) is aligned with the inertial frame $\mathbf{N}$, the magnetic vector $\mathbf{M}$ accounted by the sensor solved in the inertial frame $\mathbf{N}$ is given by,

$$\mathbf{M}_{\mathbf{N}} = B \begin{bmatrix} \cos \delta \\ 0 \\ \sin \delta \end{bmatrix}, \quad (10)$$

where B is the intensity of the Earth magnetic field, and δ is the magnetic inclination. Both quantities, B and δ , depend on the spatial location on the Earth surface. Following the same procedure used to obtain (9), the magnetic vector $\mathbf{M}$ for an arbitrary orientation of the

platform, expressed in the body-fixed frame $\mathbf{B}$, can be expressed as follows,

$$\mathbf{M}_B = \begin{bmatrix} M_\xi \\ M_\eta \\ M_\zeta \end{bmatrix} = \mathbf{T}_{BN} \begin{bmatrix} M_x \\ 0 \\ M_z \end{bmatrix} = \begin{bmatrix} (q_0^2 + q_1^2 - \frac{1}{2}) M_x + (q_1 q_3 - q_0 q_2) M_z \\ (q_1 q_2 - q_0 q_3) M_x + (q_0 q_1 + q_2 q_3) M_z \\ (q_0 q_2 + q_1 q_3) M_x + (q_0^2 + q_3^2 - \frac{1}{2}) M_z \end{bmatrix}. \quad (11)$$

where $M_x = B \cos \delta$ and $M_y = B \sin \delta$.

Finally, we obtain the outputs of the system, namely the acceleration vector in (9) and the magnetic vector in (11), as a function of the quaternion $\mathbf{q}$, expressed in the body-fixed frame. These equations along with the highly nonlinear kinematic differential equation derived in (6) give the necessary structure to build a nonlinear observer.

3. Proposed estimation strategy based on a nonlinear observer

A new strategy for estimating the quaternions is proposed in this work. With this aim, a nonlinear Luenberger observer is designed from the nonlinear model (6), which can be expressed as follows,

$$\begin{aligned} \dot{\mathbf{q}} &= \frac{1}{2} \tilde{\omega}_{(t)} \mathbf{q} \\ \mathbf{y} &= \mathbf{h}(\mathbf{q}) \end{aligned}$$

where $\mathbf{y}$ is the output (measurement) vector, obtained as a nonlinear function $\mathbf{h}(\mathbf{q})$ of the state vector $\mathbf{q}$.

For this system, the structure of the proposed observer is given by [31],

$$\dot{\hat{\mathbf{q}}} = \frac{1}{2} \tilde{\omega}_{(t)} \hat{\mathbf{q}} + \left(\frac{\partial \Psi(\hat{\mathbf{q}})}{\partial \hat{\mathbf{q}}} \right)^{-1} \mathbf{G} (\mathbf{y} - \mathbf{h}(\hat{\mathbf{q}})) \quad (12)$$

where $\hat{\mathbf{q}}$ is the estimation of the state vector, $\Psi(\hat{\mathbf{q}})$ is a nonlinear transformation that allows us to separate the states of the system into a system with an invariant linear part and a nonlinear part, and $\frac{\partial \Psi(\hat{\mathbf{q}})}{\partial \hat{\mathbf{q}}}$ is the Jacobian matrix. $\mathbf{G}$ is the observer gain matrix whose elements are calculated to obtain the desired convergence speed.

The nonlinear transformation $\Psi(\hat{\mathbf{q}})$ can be designed by using the outputs of the system and their derivatives as detailed in [31]. In the problem presented in this paper there are 6 measurements (a_ξ , a_η , a_ζ , M_ξ , M_η , M_ζ) which could be used to construct the observer. However, the system given by (6) only requires 4 measurements, which allows us to build $\Psi(\hat{\mathbf{q}})$ of different ways using the most convenient measurements combinations.

Initially, the following 4 measurements were chosen in order to design the observer,

$$\mathbf{h}(\hat{\mathbf{q}}) = \mathbf{h}_1(\hat{\mathbf{q}}) = \begin{bmatrix} a_\xi(\hat{\mathbf{q}}) \\ a_\eta(\hat{\mathbf{q}}) \\ a_\zeta(\hat{\mathbf{q}}) \\ M_\xi(\hat{\mathbf{q}}) \end{bmatrix} \quad (13)$$

In this case, the nonlinear transformation $\Psi(\hat{\mathbf{q}})$ is designed with the same components as the output estimation vector (13), that is,

$$\Psi_1(\hat{\mathbf{q}}) = \begin{bmatrix} a_\xi(\hat{\mathbf{q}}) \\ a_\eta(\hat{\mathbf{q}}) \\ a_\zeta(\hat{\mathbf{q}}) \\ M_\xi(\hat{\mathbf{q}}) \end{bmatrix} \quad (14)$$

To implement the observer, it is necessary to calculate the matrix inverse $\left(\frac{\partial \Psi_1(\hat{\mathbf{q}})}{\partial \hat{\mathbf{q}}} \right)^{-1}$ (see Appendix) and ensure that it will exist for the domain of possible measurements. Indeed, the determinant of $\frac{\partial \Psi(\hat{\mathbf{q}})}{\partial \hat{\mathbf{q}}}$ should not be null for any measurement combinations.

Given that there are 6 available measurements (and only 4 are used at a time), different groups of measurements can be chosen as outputs. It can be shown that there are no two Jacobian matrices build from

different measurements being singular at the same time. Therefore, to avoid singularities, the following strategy is implemented: when the Jacobian matrix of $\Psi(\hat{\mathbf{q}}) = \Psi_1(\hat{\mathbf{q}})$ is near to be singular, another group of measurements is used to obtain a different transformation $\Psi(\hat{\mathbf{q}})$.

In this sense when $\frac{\partial \Psi_1(\hat{\mathbf{q}})}{\partial \hat{\mathbf{q}}}$ is near to be singular it is proposed to use the following transformation $\Psi(\hat{\mathbf{q}}) = \Psi_2(\hat{\mathbf{q}})$ and the output estimation vector $\mathbf{h}_2(\hat{\mathbf{q}})$:

$$\Psi_2(\hat{\mathbf{q}}) = \begin{bmatrix} a_\xi(\hat{\mathbf{q}}) \\ a_\eta(\hat{\mathbf{q}}) \\ a_\zeta(\hat{\mathbf{q}}) \\ M_\eta(\hat{\mathbf{q}}) \end{bmatrix} \quad \mathbf{h}(\hat{\mathbf{q}}) = \mathbf{h}_2(\hat{\mathbf{q}}) = \begin{bmatrix} a_\xi(\hat{\mathbf{q}}) \\ a_\eta(\hat{\mathbf{q}}) \\ a_\zeta(\hat{\mathbf{q}}) \\ M_\eta(\hat{\mathbf{q}}) \end{bmatrix} \quad (15)$$

The practical criteria adopted to choose which transformation will be used at a given operating point is to take the transformation whose Jacobian determinant is larger (in absolute value). The pseudo-code implemented to solve this issue is the following,

$$\begin{aligned} &\text{if } \left(\left| \frac{\partial \Psi_1(\hat{\mathbf{q}})}{\partial \hat{\mathbf{q}}} \right| > \left| \frac{\partial \Psi_2(\hat{\mathbf{q}})}{\partial \hat{\mathbf{q}}} \right| \right) \\ &\quad \Psi(\hat{\mathbf{q}}) = \Psi_1(\hat{\mathbf{q}}) \\ &\quad \mathbf{h}(\hat{\mathbf{q}}) = \mathbf{h}_1(\hat{\mathbf{q}}) \\ &\text{else} \\ &\quad \Psi(\hat{\mathbf{q}}) = \Psi_2(\hat{\mathbf{q}}) \\ &\quad \mathbf{h}(\hat{\mathbf{q}}) = \mathbf{h}_2(\hat{\mathbf{q}}) \\ &\text{end if} \end{aligned} \quad (16)$$

The gain matrix $\mathbf{G}$ was chosen in order to get an adequate convergence rate. The desired convergence time may vary according to the application. In order to select the appropriate gain, it must be taken into account that the convergence time cannot exceed the measuring bandwidth of the slowest sensor (in this case, 10 Hz for the magnetometer).

3.1. Magnetometer HIO calibration

Although there are various magnetometer calibration strategies in the bibliography, in this paper a calibration strategy based on the ellipsoid fitting strategies found in the literature was implemented, simplified to estimate the HIO. That is, it is assumed that the magnetometer is only disturbed by HIO. The ellipsoid fitting problem then becomes a sphere fitting problem at the coordinate origin, with the HIO being the offset to the coordinate origin.

In this work, a recursive calibration of HIO was implemented in order to get an accurate and reliable magnetometer measurements. The process is based on the fact that the measurements made by the magnetometer (in the absence of external effects) should be located over a sphere surface of radius B [32] (being B the modulus of Earth magnetic field). Then, for this calibration process we start with the sphere equation,

$$(\mathbf{M}_d - \mathbf{H}_{io}) \cdot (\mathbf{M}_d - \mathbf{H}_{io}) = B^2 \quad (17)$$

where $\mathbf{M}_d$ is the vector given by the magnetometer measure with HIO offset, and $\mathbf{H}_{io}$ is a vector whose components are the HIO offsets corresponding to each axis,

$$\mathbf{M}_d = \begin{bmatrix} M_{d\xi} \\ M_{d\eta} \\ M_{d\zeta} \end{bmatrix} \quad \mathbf{H}_{io} = \begin{bmatrix} H_{io\xi} \\ H_{io\eta} \\ H_{io\zeta} \end{bmatrix}$$

From (17), the residual error (r) due to parametric variation on the k th measurement can be derived as follows [32],

$$\begin{aligned} r_{[k]} &= (M_{d\xi[k]}^2 + M_{d\eta[k]}^2 + M_{d\zeta[k]}^2) - \\ &\quad - \begin{pmatrix} M_{d\xi[k]} \\ M_{d\eta[k]} \\ M_{d\zeta[k]} \\ 1 \end{pmatrix}^T \begin{pmatrix} 2 H_{io\xi} \\ 2 H_{io\eta} \\ 2 H_{io\zeta} \\ B^2 - H_{io\xi}^2 - H_{io\eta}^2 - H_{io\zeta}^2 \end{pmatrix} \end{aligned} \quad (18)$$

In order to minimize the parametric error a Recursive Least Square (RLS) algorithm is implemented [33],

$$\begin{aligned} \mathbf{K}_{[k]} &= \frac{\mathbf{P}_{[k]} \mathbf{x}_{[k]}}{\lambda + \mathbf{x}_{[k]}^T \mathbf{P}_{[k]} \mathbf{x}_{[k]}} \\ \mathbf{P}_{[k+1]} &= (\mathbf{P}_{[k]} - \mathbf{K}_{[k]} \mathbf{x}_{[k]}^T \mathbf{P}_{[k]}) \frac{1}{\lambda} \\ \hat{\theta}_{[k]} &= \hat{\theta}_{[k-1]} + \mathbf{K}_{[k]} (y_{[k]} - \mathbf{x}_{[k]}^T \hat{\theta}_{[k-1]}) \end{aligned} \quad (19)$$

where $\mathbf{P}$ and $\mathbf{K}$ are variables defined by the algorithm, λ is the forgetting factor, $\hat{\theta}_{[k]}$ and $y_{[k]}$ are formed as follow

$$\begin{aligned} \hat{\theta}_{[k]} &= \begin{bmatrix} \hat{H}_{\text{io}\xi_{[k]}} \\ \hat{H}_{\text{io}\eta_{[k]}} \\ \hat{H}_{\text{io}\zeta_{[k]}} \\ 1 \end{bmatrix} \\ y_{[k]} &= M_{\xi_{[k]}}^2 + M_{\eta_{[k]}}^2 + M_{\zeta_{[k]}}^2 \end{aligned} \quad (20)$$

The vectors and matrices dimensions must meet,

$$\begin{aligned} \dim[\mathbf{x}_{[k]}] &= \dim[\mathbf{K}_{[k]}] = \dim[\hat{\theta}_{[k]}] = 4 \times 1 \\ \dim[\mathbf{P}_{[k]}] &= 4 \times 4 \\ \dim[\lambda] &= \dim[y_{[k]}] = 1 \end{aligned} \quad (21)$$

The calibration process should be carried out during the time required by the recursive estimator to converge, which is defined as T_0 . Once the recursive calibration process is finished, in a practical implementation the offset values obtained after T_0 should be stored to be used for the orientation estimation,

$$\begin{bmatrix} \hat{H}_{\text{io}\xi} \\ \hat{H}_{\text{io}\eta} \\ \hat{H}_{\text{io}\zeta} \end{bmatrix} = \begin{bmatrix} \hat{H}_{\text{io}\xi_{[T_0]}} \\ \hat{H}_{\text{io}\eta_{[T_0]}} \\ \hat{H}_{\text{io}\zeta_{[T_0]}} \end{bmatrix} \quad (22)$$

Finally, the magnetometer measurements free of HIO are obtained as the difference between,

$$\begin{bmatrix} M_{\xi_{[k]}} \\ M_{\eta_{[k]}} \\ M_{\zeta_{[k]}} \end{bmatrix} = \begin{bmatrix} M_{\xi_{[k]}} - \hat{H}_{\text{io}\xi} \\ M_{\eta_{[k]}} - \hat{H}_{\text{io}\eta} \\ M_{\zeta_{[k]}} - \hat{H}_{\text{io}\zeta} \end{bmatrix} \quad (23)$$

In order to obtain a good HIO offset estimation the recursive calibration process must be persistently excited with different measurements (e.g. rotate the board in different directions during the calibration process), ensuring the RLS algorithm convergence.

3.2. Digital implementation

The discretization of the system is carried out by using the forward-Euler method with a sampling rate of 50 Hz ($T_s = 0.02$ s). A 50 Hz sampling frequency was chosen so as to be an order above the bandwidth of the system sensors. This frequency was found adequate for the application, but it could be even lower to reduce the computational burden, being careful in its selection due to the convergence of the numerical method.

Then, the following difference equation obtained from the discretization of (12) is implemented in the microcontroller,

$$\hat{\mathbf{q}}_{[k+1]} = \hat{\mathbf{q}}_{[k]} + \left[\frac{1}{2} \tilde{\omega}_{[k]} \hat{\mathbf{q}}_{[k]} + \left(\frac{\partial \Psi(\hat{\mathbf{q}}_{[k]})}{\partial \hat{\mathbf{q}}_{[k]}} \right)^{-1} \mathbf{G} (y_{[k]} - \mathbf{h}(\hat{\mathbf{q}}_{[k]})) \right] T_s \quad (24)$$

Here, the dimensions of vectors and matrices are the following,

$$\begin{aligned} \dim[\hat{\mathbf{q}}_{[k]}] &= \dim[\hat{\mathbf{q}}_{[k+1]}] = \dim[y_{[k]}] = \dim[\mathbf{h}(\hat{\mathbf{q}}_{[k]})] = 4 \times 1 \\ \dim[\tilde{\omega}_{[k]}] &= \dim[\mathbf{G}] = \dim\left[\left(\frac{\partial \Psi(\hat{\mathbf{q}}_{[k]})}{\partial \hat{\mathbf{q}}_{[k]}}\right)^{-1}\right] = 4 \times 4 \\ \dim[T_s] &= 1 \end{aligned}$$

In the digital implementation of the proposed method is where the quaternions show important advantages. Once the inverse matrix

Table 1
MPU9150 main technical characteristics.

Gyro Range °/s	Gyro Rate Noise dps/√Hz	Accel range g	Accel Sensitivity LSB/g	ADC bits
±250	0.005	±2	±1200	16
±500	"	±4	"	16
±1000	"	±8	"	16
±2000	"	±16	"	16
Magnetometer range μT				ADC bits
±1200				13

is obtained, the remaining calculus for implementing (24) are additions and products, without the need of calculating trigonometric functions (which need more CPU computational time or more memory for look-up tables), as proposed in other implementations.

Although there are some previous implementations of orientation estimation for platforms which use quaternions, most of them use Kalman filters or extended Kalman filters. All these strategies require the numerical calculation of an inverse matrix at each sampling time, which cannot be previously precalculated.

In the proposed estimation strategy, the main difficulty to implement (24) in a microcontroller is the calculation of the inverse of the jacobian matrix. For this reason, the analytical form of the jacobian was previously obtained so as to simplify its programming on the microcontroller (see Appendix).

It must be mentioned that the magnetometer bandwidth is not programmable and it is not clearly expressed in the sensor data sheet. Therefore, it was experimental obtained, obtaining a 10 Hz update frequency. The bandwidth of the accelerometer and gyroscope included in the MPU9150 sensor can be programmed through internal registers, and it was set to 25 Hz. Both measurements were filtered by first-order digital low-pass filters implemented in the microcontroller, in order to standardize its 10 Hz bandwidth with the magnetometer, because for the implementation of the observer it is important that all the measurements have the same bandwidth.

The gain $\mathbf{G}$ was chosen so that the observer's bandwidth is 10 Hz, according to the bandwidth of the available sensors. This gain could be modified if it were desired to decrease the observer's bandwidth for a particular application.

$$\mathbf{G} = \begin{pmatrix} 63 & 1 & 0 & 0 \\ 0 & 63 & 1 & 0 \\ 0 & 0 & 63 & 1 \\ 0 & 0 & 0 & 63 \end{pmatrix} \quad (25)$$

4. Experiments

A laboratory prototype (LP) was built in order to verify experimentally the proposed estimation strategy. This prototype is composed by a base board (microcontroller programming kit) which includes the microcontroller chip, additional electronics for its operation, programming interface, connection to the PC and a connector for other boards. In this connector a board containing the inertial sensor MPU9150 is placed. See sensor features in Table 1 and the implemented prototype in Fig. 2.

The sensor measurements are acquired through an I²C serial bus and it is synchronized with a periodic external interrupt (data ready pin on MPU9150 sensor). After reading the sensor measurements the microcontroller routine call the observer algorithm and the HIO calibration algorithm. The observer needs about 700 [μs] to be executed which gives a 4% of CPU utilization over the 20 [ms] window available (100 MIPS ARM M4F CPU, 1.25 MIPS/Hz at 80 MHz). Both, the measurements and the variables estimated by the observer were



Fig. 2. Implemented prototype.

acquired through a virtual COM port between a computer and the microcontroller kit.

In this work, three experiments have been carried out which allow validating both the HIO estimator and the orientation observer on the LP. The objective of each experiment and the methodology are detailed below.

4.1. Experiment 1

This experiment is performed in order to calibrate the magnetometer HIO and to verify the consistency of these results. Only the LP is used and no additional elements are needed. For the calibration to be successful, the LP board should be rotated in different angles in order to excite the recursive algorithm with enough information. As it is well-known, stable HIO values and estimator convergence can be obtained by using a given set of angles (not necessary the same) about each rotation axis. The calibration routine is finished once continuous HIO levels are obtained on all three axes. To verify the coherence of the obtained HIO levels, it can be considered that the offset values converge to constant values, and the calculation of the Earth's magnetic field is consistent with the location on which the tests were carried out while the estimator continued running.

4.2. Experiment 2

This experiment is performed in order to verify the convergence and the stability of the observer. A flat horizontal platform with a known position and orientation is used to start the experiment. Then, different rotations of the LP are carried out about the three rotation axis. To test the stability and convergence of the observer (without special interest on its precision), it is necessary to provide a set of rotations of significant magnitude (not necessarily the same, as it was mentioned in experiment 1) about each axis, thus guaranteeing a wide operation range for the observer. It is important that both cases of (16) are covered ($|\frac{\partial \Psi_1(\hat{q})}{\partial \hat{q}}| > |\frac{\partial \Psi_2(\hat{q})}{\partial \hat{q}}|$ and $|\frac{\partial \Psi_1(\hat{q})}{\partial \hat{q}}| < |\frac{\partial \Psi_2(\hat{q})}{\partial \hat{q}}|$), to verify that the observer is stable and converges for both matrices ($\frac{\partial \Psi_1(\hat{q})}{\partial \hat{q}}$ and $\frac{\partial \Psi_2(\hat{q})}{\partial \hat{q}}$) and that non-smooth transitions do not occur under these conditions in the estimated variables. At the beginning of this experiment, the HIO must have been previously calibrated.

4.3. Experiment 3

The aim of the third experiment is to verify the correct estimation of the orientation. The quaternion-based strategy for estimating the orientation of a non-accelerated platform is validated by comparing predicted numerical results against experimental data. The mechanism used to carry out the experimental measurements consist of a two degree of freedom milling head. As can be observed in Fig. 3, it was modified by adding a non-metallic extensor arm in order to avoid magnetic interference.

The configuration of the mechanism shown in Fig. 3 is specified by using two angular coordinates. The angle ϕ takes into account the rotation about the unit vector $\hat{n}_1$, whereas the angle ψ accounts for the rotation about the body-fixed axis $\hat{b}_3$. The orientation of the milling head with respect to the inertial frame, N , can be easily described by using a rotation representation based on Euler angles. In this work, a 1–2–3 rotation sequence in terms of the angles ϕ , θ and ψ was adopted. Because the milling head is constrained to rotate about the $\hat{a}_2$ axis, the θ angle, in the 1–2–3 Euler's sequence, must be zero for all time. The function that maps the vector of Euler angles to its rotation matrix can be mathematically stated as follows,

$$\mathbf{R}_{123}(\phi, \theta, \psi) = \mathbf{R}_1(\phi) \mathbf{R}_2(\theta) \mathbf{R}_3(\psi) = \mathbf{R}_{NB},$$

where

$$\mathbf{R}_{123}(\phi, \theta, \psi) = \mathbf{R}_{NB} : {}^3 \rightarrow SO(3)$$

and it has the following form,

$$\mathbf{R}_{NB} = \begin{bmatrix} c\psi c\theta & -s\psi c\theta & s\theta \\ s\psi c\phi + c\psi s\theta s\phi & c\psi c\phi - s\psi s\theta s\phi & -c\theta s\phi \\ s\psi s\phi - c\psi s\theta c\phi & c\psi s\phi + s\psi s\theta c\phi & c\theta c\phi \end{bmatrix},$$

being $s := \sin$ and $c := \cos$.

Finally, the inverse mapping $\mathbf{h}_{123}$ from unit quaternions to Euler angles can be straightforward obtained by comparing the rotation tensors derived in (3) and (13), i.e. $\mathbf{T}_{NB}(\mathbf{q})$ and $\mathbf{R}_{NB}$, in a component-wise level [34]. After some algebraic manipulations, the inverse mapping $\mathbf{h}_{123}(\mathbf{T}_{NB}(\mathbf{q})) : SO(3) \rightarrow {}^3$ is given by,

$$\mathbf{h}_{123}(\mathbf{T}_{NB}(\mathbf{q})) = \begin{Bmatrix} \phi \\ \theta \\ \psi \end{Bmatrix} = \begin{Bmatrix} -\tan^{-1} \left(\frac{q_2 q_3 - q_0 q_1}{q_0^2 + q_3^2 - \frac{1}{2}} \right) \\ \sin^{-1} (q_1 q_3 + q_0 q_2) \\ -\tan^{-1} \left(\frac{q_1 q_2 - q_0 q_3}{q_0^2 + q_1^2 - \frac{1}{2}} \right) \end{Bmatrix}. \quad (26)$$

For computation purposes, the function $\tan^{-1}$ and $\sin^{-1}$ are replaced by the four-quadrant inverse tangent function and the inverse sine function, namely,

$$\mathbf{h}_{123}(\mathbf{T}_{NB}(\mathbf{q})) = \begin{Bmatrix} \phi \\ \theta \\ \psi \end{Bmatrix} = \begin{Bmatrix} -\text{atan2} \left(q_2 q_3 - q_0 q_1, q_0^2 + q_3^2 - \frac{1}{2} \right) \\ \text{asin} (q_1 q_3 + q_0 q_2) \\ -\text{atan2} \left(q_1 q_2 - q_0 q_3, q_0^2 + q_1^2 - \frac{1}{2} \right) \end{Bmatrix}. \quad (27)$$

5. Results and discussion

The obtained experimental results are presented in three parts, which correspond to each of the experiments described in the previous section. .

5.1. Results from experiment 1: HIO calibration

First, Section 5.1 shows the HIO estimation through the convergence to constant offset values and the calculation of the Earth's magnetic field consistent with the location on which the tests were carried out.

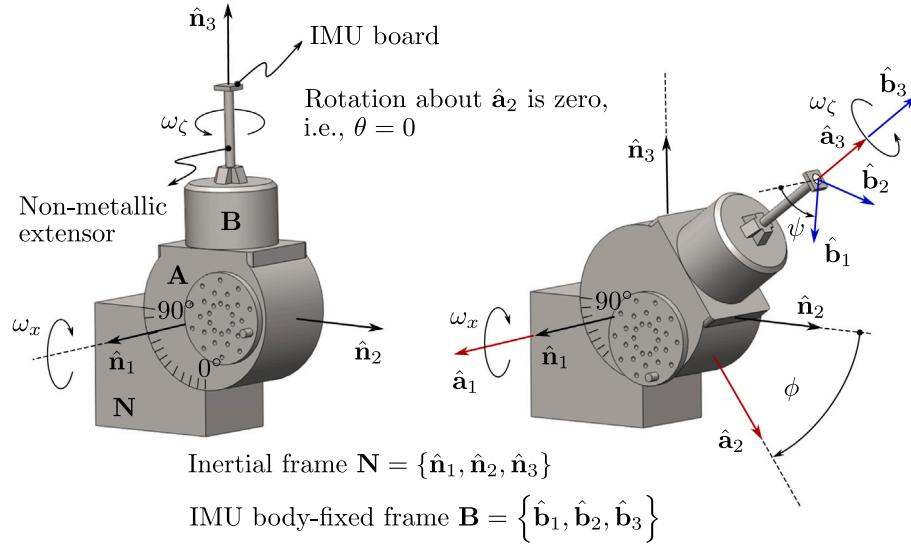
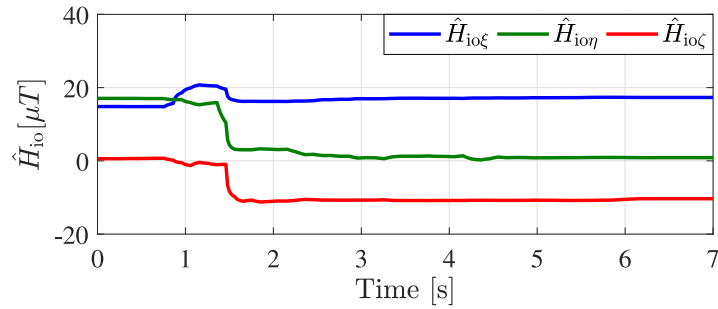
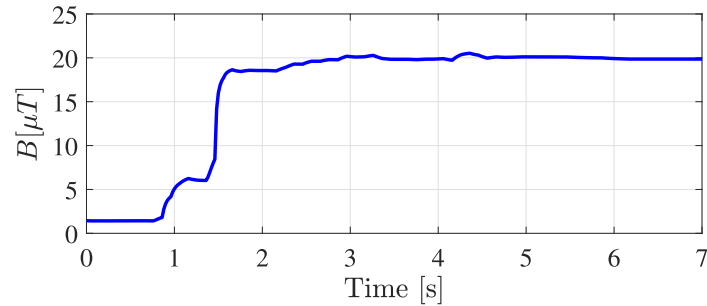


Fig. 3. Mechanism used to carry out the experimental measurement.



(a)



(b)

Fig. 4. Calibration procedure results: (a) HIO estimation; (b) $|B|$.

The calibration routine is initiated when the system is turned on, and it lasts seven seconds. During this time, the user must rotate the sensor board in different angles in order to excite the recursive algorithm with enough information.

To demonstrate the behavior of the proposed calibration algorithm, the following test was performed: the calibration process is started at $t = 0$ s. At $t = 0.5$ s the board is manually rotated $\pm 45^\circ$ first in b_1 , then $\pm 45^\circ$ in b_2 , and finally $\pm 45^\circ$ in b_3 . This sequence ends at $t = 2$ s and is repeated periodically up to the end of the calibration process ($t = 7$ s), as shown in Fig. 4.

The result of calibration is presented in Fig. 4(a), where the estimation of the three offset components ($H_{10\xi}$, $H_{10\eta}$, $H_{10\zeta}$) are shown. It can

be seen that the estimated offset values for these three variables reach a constant level from the instant $t = 1.5$ s, regardless of whether the board continues rotating. This is an expected result since the HIO levels are constant, independent of the orientation of the plate. Therefore, it can be concluded that the calibration process take less than 2 s. After that time, both B and $\mathbf{H}_{10}$ take the steady state values even if the user continues rotating the sensor.

In order to verify the validity of the obtained results, the modulus of the Earth magnetic field is calculated from (17) using the estimated values of $\mathbf{H}_{10}$. These results are shown in Fig. 4(b), where it can be seen that the steady state value of $|B|$ is about 19 μT which match with the value on the magnetic letters for this geographical location

(central Argentina). In a different location around the world, this value can be higher than $50 \mu\text{T}$ [35]. It can be noticed that the HIO values are high (in the same order of $|B|$), highlighting the importance of this offsets estimation to obtain useful magnetometer measures, in order to be used later by the observer. Once this routine finalizes (when the HIO estimator converge), the observer starts working, while the calibration routine continues running in parallel, in order to take into account the situations in which variations in the HIO occur.

5.2. Results from experiment 2: Dynamic test

Once the device is calibrated using the procedure shown in the previous subsection, in order to verify the convergence of the proposed observer and to analyze the variables involved in the estimation process, a series of rotations movements were performed as described next.

First the LP is placed in a known position: horizontal and 30 degrees orientation respect to the north. At $t = 10$ s it is rotated around $\hat{\mathbf{b}}_3$ axis (see Fig. 1), later 75 and -75 degrees around $\hat{\mathbf{b}}_2$ axis, subsequently the same movement (75 and -75 degrees) around $\hat{\mathbf{b}}_1$ axis, and finally around the $\hat{\mathbf{b}}_3$ axis by returning to the original (starting) position.

In Fig. 5(a) the estimated quaternion is shown for this test. The figure does not give a physical meaning that can be described in a simple view, but it is shown due to it is the state vector from which the Euler angles (Fig. 5(b)) were obtained using (27). As can be seen, the estimated Euler angles coincide with the movements described in this test.

Also, to verify the observer performance, Figs. 5(c) and 5(d) show the 6 measurements and their respective estimations, where it can be seen that the estimation errors are small respect to the corresponding measurements magnitude.

Fig. 5(e) shows the evolution of the Jacobian matrix determinants corresponding to the transformations $\Psi_1(\hat{q})$ and $\Psi_2(\hat{q})$, where it can be seen that the determinants are not null simultaneously. Then, as it is proposed, the observer uses the transformation which determinant is greater (in absolute value) (see pseudo-algorithm (16)).

In order to verify that the magnetometer measurements (calibrated) are correct, the instantaneous modulus of the magnetic field vector is calculated as follows,

$$\frac{B_i[k]}{|B|} = \frac{\sqrt{(M_{dx[k]} - H_{iox})^2 + (M_{dy[k]} - H_{ioy})^2 + (M_{dz[k]} - H_{ioz})^2}}{|B|} \quad (28)$$

This equation uses the 3 magnetometer measurements (removing the HIO). If the magnetometer calibration process were perfect and there were no noise in the measurements, the value obtained by (28) would be 1, because the vector is constant in a geographical location. Since in this work only HIO is removed, small variation around 1 could be expected, as can be appreciated in Fig. 6.

It was also verified that the execution of the HIO estimation algorithm and the observer needs $700 \mu\text{s}$. This means that considering a sampling frequency of 50 Hz, this represents 3.5 percent of the time for each sampling period.

5.3. Results from experiment 3: observer validation

In order to validate the estimation of the proposed observer, a -30 degrees rotation of the milling head was made around the axis $\hat{\mathbf{a}}_1$ ($\phi = -30^\circ$). After this, successive rotations of the milling head were made around the axis $\hat{\mathbf{b}}_3$ at 18 degrees each using the graduated scale of the device, completing one revolution. That is, 20 points were recorded in order to complete a full 360 degrees turn.

During the test, the values obtained from the estimated quaternions, $\hat{q}$, were recorded for each of these points and transformed using (27). The obtained estimation angles are shown in Fig. 7. As it can be seen, $\hat{\psi}$ follows ψ during the 360 degrees of rotation with a small error. It

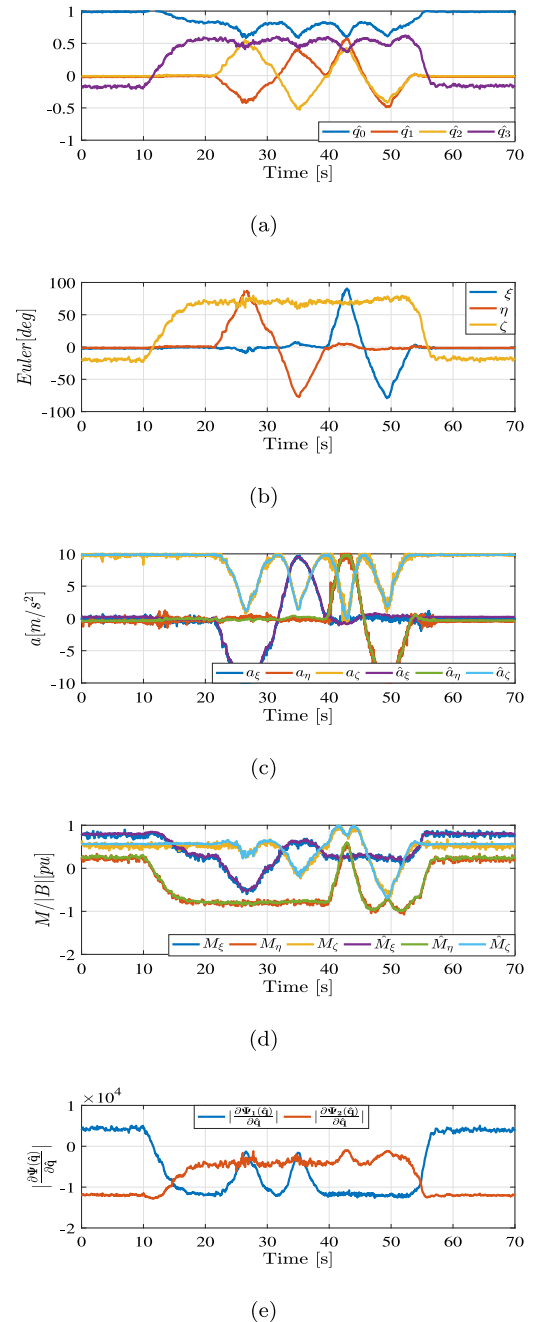


Fig. 5. Dynamic test: (a) estimated quaternion; (b) Euler orientation angles estimation; (c) measured and estimated accelerations; (d) measured and estimated magnetic field; (e) Jacobian matrix determinants.

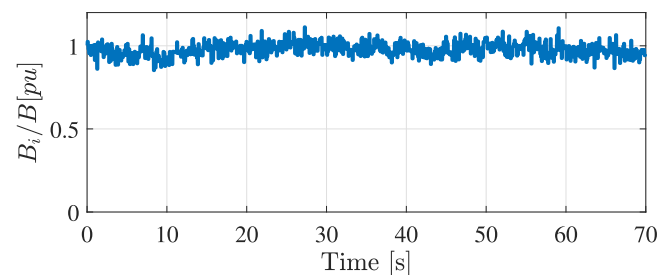


Fig. 6. Instantaneous magnetic field B_i in per unity.

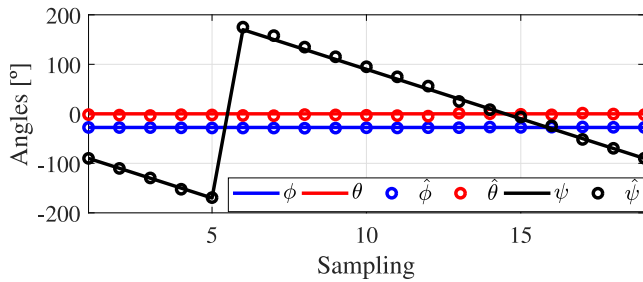


Fig. 7. Observer validation.

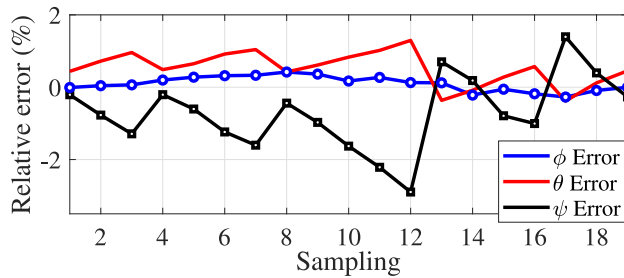


Fig. 8. Observer errors.

can also be appreciated that $\hat{\phi}$ and $\hat{\theta}$ always remain at the same value, and it coincides with the initial values of the experiment. Finally, in Fig. 8 the estimation error can be appreciated, which was calculated as the difference between the actual angles set in the milling head and the estimated angles obtained with the observer.

From the results shown in the figure, the root mean square error was calculated, resulting in a 3 degrees value. This estimation error is caused by various sources, such as the measurement errors of each of the sensors, their range and their resolution (number of bits). The accelerometer and gyroscope measurement range is software configurable, therefore they were adjusted to the measurement ranges of the problem. Also, both sensors have a 16-bit resolution. On the other hand, the measurement range of the magnetometer is not adjustable and is two orders higher than the magnitude of the Earth's magnetic field (1200 μ T vs 20 μ T) and its resolution is 13 bits. This causes the maximum quantization error of the magnetometer to be approximately 0.7%, being the main source of error in the orientation estimation.

6. Conclusion

In this work the design and implementation of an IMU based estimation strategy for obtaining the orientation angles of a non-accelerated low cost measurement system was proposed. In addition, a recursively estimator was implemented to determine the magnetometer HIO.

The observer was discretized at a sampling frequency selected according to the bandwidths of the IMU sensors. Both the observer and the HIO estimator were implemented on an experimental laboratory prototype on a 32-bit microcontroller. Experimental results for the initial calibration procedure and orientation estimation demonstrate the system performance.

The convergence time of the HIO estimator was shown in a first part, calibrating in this way the HIO of the magnetometer. In a second part, once the magnetometer was calibrated, the dynamic response and the convergence of the observer was analyzed. It is concluded that the observer converges to the expected values and that the estimation errors of the observer output variables are small. In addition, it is verified that the instantaneous value of $|B|$ is consistent, allowing to verify that the previous calibration was performed correctly. Finally, it is also demonstrated that the computational cost of the implementation of the

observer and the estimator is less than 5 percent of the capacity of the microcontroller. Therefore, although the selected microcontroller is of relatively low cost, in an industrial application a lower performance and lower cost microcontroller could be used.

In the last test, the laboratory prototype was placed in a drilling machine that allows rotations in two axes with a standard graduation. This allowed comparing the magnitude of the estimated orientation angles with respect to the standard rotations angles, verifying the root mean square error in a 3 degrees value. The magnitude of this error could be decreased if a magnetometer with a measurement range with higher resolution, or closer to the magnitude of the Earth's magnetic field intensity, were used. Also, a more complex calibration of the magnetometer should be performed, such as considering the SIO.

CRedit authorship contribution statement

Diego A. Aligia: Conceptualization, Data curation, Software, Validation, Writing - original draft, Investigation, Visualization, Methodology, Formal analysis. **Bruno A. Rocca:** Formal analysis, Methodology, Validation, Writing - review & editing. **Cristian H. De Angelo:** Formal analysis, Funding acquisition, Supervision, Project administration, Writing - review & editing, Visualization, Methodology. **Guillermo A. Magallán:** Writing - original draft, Resources, Validation. **Guillermo N. González:** Validation, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Analytical calculations of the transformation

To implement the proposed observer (12) the transformations defined in (14) and (15) are obtained as follows,

$$\Psi_1(\mathbf{q}) = \begin{bmatrix} a_\xi(\mathbf{q}) \\ a_\eta(\mathbf{q}) \\ a_\zeta(\mathbf{q}) \\ M_\xi(\mathbf{q}) \end{bmatrix} = \begin{bmatrix} (2q_1q_3 - 2q_0q_2)g \\ (2q_0q_1 + 2q_2q_3)g \\ (2q_0^2 + 2q_3^2 - 1)g \\ (q_0^2 + q_1^2 - \frac{1}{2})M_x + (q_1q_3 - q_0q_2)M_z \end{bmatrix}$$

$$\Psi_2(\mathbf{q}) = \begin{bmatrix} a_\xi(\mathbf{q}) \\ a_\eta(\mathbf{q}) \\ a_\zeta(\mathbf{q}) \\ M_\eta(\mathbf{q}) \end{bmatrix} = \begin{bmatrix} (2q_1q_3 - 2q_0q_2)g \\ (2q_0q_1 + 2q_2q_3)g \\ (2q_0^2 + 2q_3^2 - 1)g \\ (q_1q_2 - q_0q_3)M_x + (q_0q_1 + q_2q_3)M_z \end{bmatrix}$$

Their Jacobians results in,

$$\frac{\partial \Psi_1(\mathbf{q})}{\partial \mathbf{q}} = 2 \begin{bmatrix} -gq_2 & gq_3 & -gq_0 & gq_1 \\ gq_1 & gq_0 & gq_3 & gq_2 \\ gq_0 & -gq_1 & -gq_2 & gq_3 \\ M_xq_0 - M_zq_2 & M_xq_1 + M_zq_3 & -M_xq_2 - M_zq_0 & M_xq_3 - M_zq_2 \end{bmatrix} \quad (29)$$

$$\frac{\partial \Psi_2(\mathbf{q})}{\partial \mathbf{q}} = 2 \begin{bmatrix} -gq_2 & gq_3 & -gq_0 & gq_1 \\ gq_1 & gq_0 & gq_3 & gq_2 \\ gq_0 & -gq_1 & -gq_2 & gq_3 \\ -M_xq_3 + M_zq_1 & M_xq_2 + M_zq_0 & M_xq_1 + M_zq_3 & -M_xq_0 - M_zq_2 \end{bmatrix} \quad (30)$$

The inverse of (29) and (30) were analytically calculated, but are omitted here due to space constraints. Therefore, in the microcontroller

matrices $\left(\frac{\partial \psi_1(q)}{\partial q}\right)^{-1}$ and $\left(\frac{\partial \psi_2(q)}{\partial q}\right)^{-1}$ are evaluated for implementing the observer. That is to say, the inverse operation of a matrix is not solved in the microcontroller, but the analytical expressions of the inverse matrix are evaluated.

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